

DYLAN LAPLACE MERMOUD

Postdoctoral researcher in game theory, quantum algorithms and algebraic combinatorics
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PERSONAL INFORMATION

Born in Saint-Michel (16), France, on October 25, 1996. French nationality.

AREAS OF INTEREST

Discrete mathematics, cooperative game theory, combinatorial optimization, quantum algorithms, Coxeter groups, algebraic combinatorics, differential equations on combinatorial structures.

EMPLOYMENT

Postdoctoral researcher Oct. 2023 – today

- ▷ Unité de Mathématiques Appliquées, ENSTA, Institut Polytechnique de Paris
- ▷ Secondary affiliation: CEDRIC laboratory, Conservatoire National des Arts et Métiers
- ▷ Supervisors: ANDREA SIMONETTO and SOUROUR ELLOUMI

Teaching and research associate Sept. 2022 – Aug. 2023

- ▷ Centre d'Économie de la Sorbonne, Université Paris I Panthéon-Sorbonne

Doctoral contract employee Sept. 2019 – Aug. 2022

- ▷ Centre d'Économie de la Sorbonne, Université Paris I Panthéon-Sorbonne

EDUCATION

Doctoral degree, Mathematics, *Université Paris I Panthéon-Sorbonne* Sept. 2019 – Sept. 2023

- ▷ Title: “[Geometry of Set Functions in Game Theory: Combinatorial and Computational Aspects](#)”
- ▷ Supervisors: MICHEL GRABISCH and PETER SUHDÖLTER
- ▷ Reviewers: TAMÁS SOLYMOSI and MARINA NUÑEZ, president of the jury: PHILIPPE BICH, jury members: JEAN-JACQUES HERINGS and PHILIPPE SOLAL

Master degree, Mathematics, *Sorbonne Université* Sept. 2017 – June 2019

- ▷ Jointly accredited by École Polytechnique and École Nationale des Ponts et Chaussées.

Bachelor degree, Mathematics, *Sorbonne Université* Sept. 2014 – June 2017

PUBLICATIONS, PROCEEDINGS, AND PREPRINTS

Publications

11. The algebraic structures of social organizations: the operad of cooperative games
LAPLACE MERMOUD, D., ROCA I LUCIO, V.
[Algebraic Combinatorics](#) • 2026 • 49 pages • [arXiv.2507.01969v2](#)

10. Variational quantum algorithms for permutation-based combinatorial problems: Optimal ansatz generation with applications to quadratic assignment problems and beyond
LAPLACE MERMOUD, D., SIMONETTO, A., ELLOUMI, S.
[Quantum](#), vol. 10, p. 1998 • 2026 • 36 pages • [arXiv.2509.05981v3](#)
9. Attraction of the core and the cohesion flow
LAPLACE MERMOUD, D.
[Theory and Decision](#), vol. 99, pp. 377–392 • 2025 • 15 pages • [arXiv.2507.02918](#)
8. Minimal balanced collections and their application to core stability and other topics of game theory
LAPLACE MERMOUD, D., GRABISCH, M., SUDHÖLTER, P.
[Discrete Applied Mathematics](#), vol. 341, pp. 60–81 • 2023 • 32 pages • [arXiv.2507.05898](#)

Proceedings

7. Scaling QAOA: transferring optimal adiabatic schedules from small-scale to large-scale variational circuits
NZONGANI, U., LAPLACE MERMOUD, D., BRAIDA, A.
[2026 IEEE Conference on Quantum Control, Computing and Learning](#) • 2026 • 6 pages • [arXiv.2602.14986](#)
6. Combinatorics on social configurations
LAPLACE MERMOUD, D., POPOLI, P.
[Electronic Proceedings in Theoretical Computer Science](#), vol. 403, pp. 128–133 • 2024 • 6 pages

Preprints

5. On the closest balanced game
GARCIA-SEGADOR, P., GRABISCH, M., LAPLACE MERMOUD, D., MIRANDA, P.
[arXiv preprint arXiv.2601.15318](#) (Under review) • 2025 • 31 pages
4. Sampled-based guided quantum walk: Non-variational quantum algorithm for combinatorial optimization
NZONGANI, U., LAPLACE MERMOUD, D., DI MOLFETTA, G., SIMONETTO, A.
[arXiv preprint arXiv.2509.15138](#) (Revised and resubmitted) • 2025 • 27 pages
3. Diffusion in dynamic networks with time-varying inputs to allocate responsibility
VAN DEN ENDE, R., LAPLACE MERMOUD, D.
[arXiv preprint arXiv.2301.02728](#) • 2025 • 25 pages
2. Projection onto the core: An optimal reallocation to correct market failure
LAPLACE MERMOUD, D.
[arXiv preprint arXiv.2411.11810](#) • 2024 • 30 pages
1. On the formation of steady coalitions
LAPLACE MERMOUD, D.
[arXiv preprint arXiv.2410.05087](#) • 2024 • 21 pages

RESEARCH VISITS

- ▷ MICHEL GRABISCH at the [Algorithmic game theory in socioeconomics \(AGATE\)](#) center
Charles University, Prague, Czech Republic • one or two weeks • 2026 (forthcoming)
- ▷ MĀRIS OZOLS and KRYSTAL GUO at the [Discrete Mathematics and Quantum Information group](#)
Universiteit van Amsterdam and CWI, The Netherlands • two weeks • 2025
- ▷ STEFAN WÖRNER at the [Quantum Computational Science group](#)
IBM Research Europe - Zurich, Suisse • one week • 2025
- ▷ VICTOR ROCA I LUCIO at the [Laboratory for Topology and Neurosciences](#)
École Polytechnique Fédérale de Lausanne (EPFL), Suisse • one week • 2023
- ▷ PIERRE POPOLI at the [Laboratory of Discrete Mathematics](#)
Université de Liège, Belgique • one week • 2023

TEACHING

Lectures

- ▷ Quantum computing 2025 – 2026
MPRO & ensIIE • Master 2 in informatics • 6h
- ▷ Game theory 2024 – 2026
ENSTA & Institut Polytechnique de Paris • Master 1 in mathematics • 7h30 • [Lecture notes](#)

Tutorials

- ▷ Quantum computing 2025 – 2026
MPRO & ensIIE • Master 2 in informatics • 15h
- ▷ Game theory 2024 – 2026
ENSTA & Institut Polytechnique de Paris • Master 1 in mathematics • 10h
- ▷ Graph theory 2023 – 2026
ENSTA & Institut Polytechnique de Paris • Master 1 in mathematics • 16h30
- ▷ Linear programming 2020 – 2023
Université Paris I Panthéon-Sorbonne • Master 1 in mathematics • 24h
- ▷ Topology and differential calculus 2022 – 2023
Université Paris I Panthéon-Sorbonne • Bachelor 3 in mathematics/informatics • 42h
- ▷ Microeconomics 2022 – 2023
Université Paris I Panthéon-Sorbonne • Bachelor 3 in mathematics/informatics • 48h
- ▷ Linear algebra 2022 – 2023
Université Paris I Panthéon-Sorbonne • Bachelor 2 in mathematics/informatics • 30h
- ▷ Linear algebra 2019 – 2023
Université Paris I Panthéon-Sorbonne • Bachelor 3 in economics • 36h
- ▷ Python programming 2019 – 2020
Université Paris I Panthéon-Sorbonne • Bachelor 1 in mathematics/informatics • 48h

SUPERVISION

- ▷ Paul-Henri Djoko (2026), Master 1 student, on reinforcement learning in game theory.
- ▷ Bruno Galibert (2025), Master 1 student, on quantum circuit simplification.
- ▷ Qui Loc Phan (2024), Master 1 student, on quantum circuit design and combinatorial optimization.

AWARDS AND FUNDINGS

- ▷ Laureate of the *Quantum-Saclay* grant, one internship funding (2025).
- ▷ Laureate of the *Passeport pour le Master* scholarship from Sorbonne Université Foundation, fully funded 5 years of study (2014-2019).

RESPONSIBILITIES

- ▷ Member of the board of directors of Sorbonne Université (2018-2020), member of the board of the mathematics department of Sorbonne Université (2018-2020).

REFEREEING

- ▷ Journal of Mathematical Economics, 4OR - A Quarterly Journal of Operations Research, Annals of Operations Research (x2), IEEE International Conference on Quantum Control, Computing and Learning (x2), CEMRACS 2025 – Quantum Computing, Operations Research Letters, Fuzzy Sets and Systems.

COMMUNICATIONS

▷ Session organization

1. “Quantum computing for combinatorial optimization” at [Programme Gaspard Monge \(PGMO\) days 2025](#), Palaiseau, France (2025).

▷ Upon invitation

2. [SIAM Conference on Optimization \(OP26\)](#), Edinburgh, United Kingdom (forthcoming, 2026).
3. 18th [Ordered Structures in Games and Decisions](#) in Michel Grabisch’s honor, Paris, France (2025).
4. 16th [Ordered Structures in Games and Decisions](#) conference, Paris, France (2023).
5. 22nd [Society for the Advancement of Economic Theory \(SAET\)](#) conference, Paris, France (2023).
6. [GO22: Games and Optimization 2022](#), Saint-Étienne, France (2022).

▷ Conferences

7. 27th [ROADEF congress](#), Tours, France (2026).
8. 20th [European Meeting on Game Theory \(SING20\)](#), Maastricht, The Netherlands (2025).
9. [International Conference in Quantum Computing \(ICoQC2025\)](#), Paris, France (2025).
10. 26th [ROADEF congress](#), Champs-sur-Marne, France (2025).
11. [Programme Gaspard Monge \(PGMO\) days 2024](#), Palaiseau, France (2024).
12. 19th [European Meeting on Game Theory \(SING19\)](#), Besançon, France (2024).
13. 13th [Random Generation of Combinatorial Structures. Polyominoes and Tiling \(GASCom\)](#), Bordeaux, France (2024).
14. 18th [European Meeting in Game Theory \(SING18\)](#), Messina, Italy (2023).
15. 26th [Coalition Theory Network \(CTN\)](#) conference, Bielefeld, Germany (2023).
16. [Programme Gaspard Monge \(PGMO\) days 2021](#), Palaiseau, France (2021).
17. 16th [European Meeting in Game Theory \(SING16\)](#), Granada (remote), Spain (2021).

▷ Seminars

18. Universiteit van Amsterdam, [Discrete Mathematics and Quantum Information](#) group seminar, Amsterdam, The Netherlands (2025).
19. IBM Research Europe - Zurich, [Quantum Optimization project](#), Zurich, Switzerland (2025).
20. Université Aix-Marseille, [CANA](#) research group seminar, Marseille, France (2025).
21. [C12](#), R&D group seminar, Paris, France (2025).
22. Université Paris Cité, [IRIF](#) quantum team seminar, Paris, France (2025).
23. École Polytechnique, [PhIQus](#) team seminar, Palaiseau, France (2025).
24. Télécom Paris, [Quriosity](#) team seminar, Palaiseau, France (2025).
25. [Quandela](#), R&D group seminar, Massy, France (2025).
26. Conservatoire National des Arts et Métiers, [Combinatorial Optimization \(OC\)](#) group seminar, Paris, France (2025).

27. ENSTA - Institut Polytechnique de Paris, [Optimization and Command \(OC\) group seminar](#), Palaiseau, France (2025).
28. Centre d'Économie de la Sorbonne, "[Mathématique Discrète, Optimisation, Décision](#)" seminar, Paris, France (2024).
29. Université de Franche-Comté, "[young researchers](#)" day, Besançon, France (2024).
30. Institut Henri Poincaré, [Junior Parisian seminar in game theory](#), Paris, France (2023).
31. ENSTA - Institut Polytechnique de Paris, [Optimization and Command \(OC\) group seminar](#), Palaiseau, France (2023).
32. Paris School of Economics, [Doctorissimes](#), Paris, France (2022).
33. Centre d'Économie de la Sorbonne, Maths department days, Paris, France (2022).
34. Centre d'Économie de la Sorbonne, "[Network and Games](#)" seminar, Paris, France (2022).
35. Centre d'Économie de la Sorbonne, [PhD seminar](#), Paris, France (2021).
36. Institut Élie Cartan, [PhD seminar](#), Nancy, France (2021).

REFERENCES

MICHEL GRABISCH

michel.grabisch@univ-paris1.fr

Honorary senior member of the Institut Universitaire de France
Professor Emeritus at Université Paris I Panthéon-Sorbonne
Head of the AGATE center and ERA chair holder at Charles University

FRÉDÉRIC MEUNIER

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President of the French Operations Research & Decision Support Society (ROADEF)
Researcher at CERMICS in the Optimization team and professor at École nationale des ponts et chaussées
Associate professor at École polytechnique

SOUROUR ELLOUMI

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Professor at ENSTA, Institut Polytechnique de Paris
Head of the Optimization and Command team at ENSTA, Institut Polytechnique de Paris
Member of the Combinatorial Optimization team at Conservatoire National des Arts et Métiers

ANDREA SIMONETTO

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Research professor at ENSTA, Institut Polytechnique de Paris
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Member of the Combinatorial Optimization team at Conservatoire National des Arts et Métiers